

CE 516: Uncertainty Analysis in Civil Engineering

Course Code	CE 516
Course Name	Uncertainty Analysis in Civil Engineering
L-T-P-C	3-0-0-3
Prerequisites	Students are suggested to have a background on basic Engineering Mathematics related course at the UG level and the design and testing aspects in Civil Engineering (CE351: Design of Reinforced Concrete Structures or, CE402: Geotechnical Engineering II, or CE401: Design of Steel Structures or, CE354P: Building and Pavement Materials Laboratory, or equivalent)
Intended for	B.Tech. 4th year/ M.S./ M.Tech./ Ph.D.
Mutual Exclusion	'None'
Approval	50th BoA (Jul 2023)

Course Contents

- **The concept of 'Risk and uncertainty' in Civil Engineering:** Difference between deterministic and uncertain parameters, sources of uncertainties in civil engineering, classification and nomenclature of uncertainties (epistemic, aleatory, parametric, etc.), the concept of hazard, vulnerability and risk, Factor of Safety, lower and upper bounds. (3 Hours)
- **Uncertainty Quantification in Civil Engineering:** Understanding the examples of discrete and continuous random variables in civil engineering such as soil data, concrete strength data, other strength and design data, uncertainty quantification of such variables in form of mean, variance, COV, histograms, percentiles, box plots, correlation, skewness, scatter, probability distributions (PDF and PMF), expectations, moments, joint probability distributions, and extreme value theories. (10 Hours)
- **Introduction to Random Process:** Introduction to earthquake ground motions as random processes, and their basic statistical, temporal, and Spatial characterization including variograms, autocorrelation functions, probability density, Interpolation methods (e.g., kriging), stationarity, ergodicity, and return period. (7 Hours)
- **Uncertainty Simulation & Propagation:** Simulating the random variables using the Monte-Carlo approach, Bootstrap method, example case-studies to understand the implications of these random input variables on the desired outcomes (e.g., structural response to random loads, case-studies from construction planning and management), derivation of the probability distributions of output variables, testing their goodness-fit, sensitivity analysis, first order second moment (FOSM) methods, and event tree analysis. (12 Hours)
- **Data interpretation and modelling:** Introduction to typical data from civil engineering experiments that needs to be modeled, identification of dependent and independent variables through example problems, Correlation analysis between the parameters in the model, identifying outliers, Functional form identification (Parametric, non-parametric), Performance evaluation, confidence interval, a brief overview on approaches to handling of missing information, Mixed-effects regression its significance and application. (6 Hours)
- **Advanced and Miscellaneous Topics:** The basics concepts of load, resistance, failure probability, factor of safety, and reliability, point estimate method, error propagation, Hasofer-Lind approach, Conditional probability, Bayes' law, Posterior distribution. (4 Hours)

Text Books

1. H-S. Ang & W.H. Tang, **Probability Concepts in Engineering: Emphasis on Applications to Civil and Environmental Engineering**, Wiley, 2006.
2. Benjamin, Jack R., and C. Allin Cornell, **Probability, statistics, and decision for civil engineers**, Courier Corporation, 2014.

References

1. Papoulis, A., **Probability, Random Variables and Stochastic Processes**, 3rd Edition, McGraw-Hill, 1991.
2. Jay L. Devore, **Probability and Statistics for Engineering and the Sciences**, Brooke & Cole, 2009.
3. Montgomery, Douglas C., and George C. Runger, **Applied statistics and probability for engineers**, John Wiley & Sons, 2007.
4. Fellin, W., Lessmann, H., Oberguggenberger, M., & Vieider, R. (Eds.), **Analyzing uncertainty in civil engineering**, Springer, 2005.
5. Ross, S. M., **Introduction to probability and statistics for engineers and scientists**, Elsevier, 2004.
6. Soong, T. T., **Fundamentals of probability and statistics for engineers**, John Wiley & Sons, 2004.